

MNIST Classification using MLE and Discriminant Analysis

Deadline: 9th Monday 11:59pm

Problem Statement

You are given MNIST (<https://www.kaggle.com/datasets/hojjatk/mnist-dataset>), which consists of handwritten digit images (0-9). Your task is to build a classification pipeline using only the digits 0, 1, and 2. Randomly select 100 samples from train set of each class leading to a total of 300 samples. Call this as train set. Randomly sample 100 samples from test set for each of those 3 classes and denote it as test set. This exercise incorporates the following concepts:

1. **Maximum Likelihood Estimation (MLE):** Estimate the mean and covariance matrix of each digit class assuming a Gaussian distribution.
2. **Discriminant Analysis for Classification:** Use Linear Discriminant Analysis (LDA) or Quadratic Discriminant Analysis (QDA) to classify test samples.

Task Description

Implement the following steps:

1. **Data Preprocessing**
 - Load the MNIST dataset and filter only classes 0, 1, and 2.
 - Convert images into feature vectors by stacking columns and normalize them to be in the range 0 to 1.
2. **Compute MLE Estimates** [CO1] [1]
 - Estimate the mean μ_c and covariance matrix Σ_c for each digit class using MLE. Use only train set. Note you need to explicitly use the equations given in the lecture. Covariance is $\frac{1}{N} \sum_i (x_i - \mu)(x_i - \mu)^\top$

- Assume the data follows a multivariate Gaussian distribution:

$$P(\mathbf{x}|y = c) = \frac{1}{(2\pi)^{d/2}|\Sigma_c|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c)\right).$$

3. Apply LDA and QDA to the test set and report the accuracies. [CO2] [2]

Your should clearly include:

- Accuracy of LDA and QDA classifiers.
- t-sne (<https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html>) plots of the train and test set.
- Discriminant value of sample test point

Accuracy Definition

Accuracy is a commonly used metric in classification tasks, measuring the proportion of correct predictions over the total number of predictions.

Mathematical Definition

Accuracy is defined as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (1)$$

If we define:

- N as the total number of test samples,
- $\hat{y}_i$ as the predicted label for the i -th sample,
- y_i as the true label for the i -th sample,

then the accuracy can be expressed as:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{K}(\hat{y}_i = y_i) \quad (2)$$

where $\mathbb{K}(\hat{y}_i = y_i)$ is an indicator function that returns 1 if the prediction is correct and 0 otherwise.

Special Cases

1. **Perfect Accuracy:** If all predictions are correct:

$$\text{Accuracy} = 1.0 \quad (100\% \text{ correct}) \quad (3)$$

2. **Worst Case Accuracy:** If no predictions are correct:

$$\text{Accuracy} = 0.0 \quad (0\% \text{ correct}) \quad (4)$$

Considerations

- Accuracy is useful when classes are balanced.
- For imbalanced datasets, accuracy can be misleading. Alternative metrics like Precision, Recall, and F1-score should be considered.